

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

|                         |                                                                                                                            |                      |                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------------------------|
| <b>Course Code:</b>     | DSA03                                                                                                                      | <b>Course Title:</b> | Natural Language Processing                    |
| <b>CO Assessed:</b>     | CO2 – Manipulation                                                                                                         | <b>Assessment:</b>   | Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING |
| <b>Weightage:</b>       | 35% of CIA                                                                                                                 |                      |                                                |
| <b>CO2 Statement:</b>   | Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3) |                      |                                                |
| <b>Student Name:</b>    | A Lokanath Reddy                                                                                                           | <b>Reg. No.:</b>     | 192425321                                      |
| <b>Section / Batch:</b> | AIML                                                                                                                       | <b>Date:</b>         |                                                |

### Code of Conduct

I A Lokanath Reddy/192425321 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

### Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

| Section / Topic                                                                                      | Focus                                                                                                                                                                                  | Q Nos |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| English Word Classes, Part of Speech Tagging                                                         | Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing. | Q1    |
| English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging                         | Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.                           | Q2    |
| Counting Words, Part of Speech Tagging, English Word Classes                                         | Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.                                     | Q3    |
| Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging | Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.                                                          | Q4    |
| English Word Classes, Rule-Based Part of Speech Tagging, Counting Words                              | Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.                                     | Q5    |

## **Annexure A – ANALYTICAL PROBLEM SOLVING**

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**1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.**

- Apply rules to decompose each word into root and suffix components.
- Differentiate between inflectional and derivational forms within the dataset.
- Normalize all variants to a common base form for efficient retrieval.
- Determine the parsed structure and normalized output for each word.
- Write a Python program to implement the above morphological analysis pipeline. The program should accept the given input words, perform decomposition, classify each suffix as inflectional or derivational, normalize the words to their common base form, and display the parsed structure and normalized output in a tabular format.

**2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.**

- Identify prefixes, suffixes, and base forms using rule-based decomposition.
- Ensure semantic consistency across derived word forms.
- Classify each transformation as inflectional or derivational.
- Produce the root and morphological breakdown for each word.
- Write a Python program to implement the above morphological parsing module. The program should accept the given input words, identify the prefixes, suffixes, and base forms, classify each transformation as inflectional or derivational, and display the morphological breakdown and normalized root word in a tabular format.

**3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.**

- Apply morphological rules to strip affixes and identify the base form.
- Handle both grammatical variations and word formation changes.
- Ensure consistent root extraction across all forms.
- Determine the stem and classification for each input word.
- Develop a Python program for a stemming-based preprocessing module used in a search engine. The program should process the input words "played", "player", "playing" by applying appropriate stemming rules to remove prefixes/suffixes where applicable, identify the stem of each word, distinguish between grammatical inflections and word-formation changes, and generate a structured output showing the original word, extracted stem, removed affix (if any), transformation type (inflectional/derivational), and the final normalized form suitable for indexing and information retrieval.

**4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.**

- Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
- Differentiate between regular and irregular inflectional patterns within the dataset.
- Ensure consistent mapping of all forms to a common root representation.
- Determine the state transitions, morphological breakdown, and normalized output for each word.

- Develop a Python program that implements a finite-state morphological parser for the input words "writes", "writing", "written". The program should construct a finite-state transition model to process regular suffixes and irregular verb forms, identify the corresponding root word, classify each input as a regular or irregular inflection, trace the sequence of state transitions during parsing, and display the state transition path, morphological components, root form, and normalized representation for each input word in a well-formatted output.

**5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.**

- Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
- Handle variations in suffix patterns while preserving semantic consistency.
- Ensure uniform root extraction across all derived forms.
- Determine the stemming steps, intermediate forms, and final stem for each input word.
- Develop a Python program that implements the Porter Stemming algorithm for the input words "relational", "relation", "relate". The program should apply the Porter stemming rules sequentially to remove derivational suffixes, display the intermediate form obtained after each stemming step, handle different suffix patterns while preserving the word meaning, and generate the final stem for each word. The output should clearly present the original word, applied stemming rule(s), intermediate forms, and the final normalized stem in a structured format suitable for document retrieval applications.

## Rubrics for Calculation

| Numerical Problems        |           |                                                                                   |                                           |                                            |                                               |
|---------------------------|-----------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Criteria                  | Weightage | Level 4 (Exemplary)                                                               | Level 3 (Proficient)                      | Level 2 (Developing)                       | Level 1 (Beginning)                           |
| Problem Understanding     | 20%       | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection          | 20%       | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       |
| Solution Process          | 25%       | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear process                              |
| Accuracy of Calculation   | 20%       | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         |
| Interpretation of Results | 15%       | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             |

| Q. No. / Criteria Level | Problem Understanding | Method Selection | Solution Process | Accuracy of Calculation | Interpretation of Results | Sub. Total | Total % |
|-------------------------|-----------------------|------------------|------------------|-------------------------|---------------------------|------------|---------|
| 1                       | 17                    | 17               | 18               | 17                      | 16                        | 85         | 85%     |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

## Introduction

Morphological analysis is the process of decomposing words into meaningful components such as roots, prefixes, and suffixes. It is useful in natural language processing applications including document indexing, sentiment analysis, search engines, text processing, and information retrieval. This report presents five rule-based problems covering morphological decomposition, derivational and inflectional analysis, stemming, finite-state parsing, and Porter stemming.

## Problem 1 – Morphological Analysis Pipeline

**Input words:** connected, connecting, connection.

The objective is to decompose each word into its root and suffix, classify the suffix as inflectional or derivational, and normalize all related forms to a common base.

| Word       | Root    | Suffix | Type         | Normalized |
|------------|---------|--------|--------------|------------|
| connected  | connect | ed     | Inflectional | connect    |
| connecting | connect | ing    | Inflectional | connect    |
| connection | connect | ion    | Derivational | connect    |

Rules:

1. -ed indicates past tense and is inflectional.
2. -ing indicates progressive form and is inflectional.
3. -ion forms a noun from a verb and is derivational.

### Python Program:

```
words = ["connected", "connecting", "connection"]
def analyze_word(word):
    if word.endswith("ing"):
        root = word[:-3]
        suffix = "ing"
        suffix_type = "Inflectional"

    elif word.endswith("ed"):
        root = word[:-2]
        suffix = "ed"
        suffix_type = "Inflectional"
    elif word.endswith("ion"):
        root = word[:-3]
        suffix = "ion"
        suffix_type = "Derivational"
    else:
        root = word
        suffix = "-"
        suffix_type = "None"
    return root, suffix, suffix_type
print("\nMORPHOLOGICAL ANALYSIS")
print("-" * 75)
print(f'{"Word":<15}{Root':<15}{Suffix':<12}{Type':<18}{Normalized':<15}")
print("-" * 75)
for word in words:
```

```

root, suffix, suffix_type = analyze_word(word)
print(f"{word:<15}{root:<15}{suffix:<12}"
      f"{suffix_type:<18}{root:<15}")

```

## Output

```

PS C:\Users\lenovo\OneDrive\Documents\DSA0308> c:: cd 'c:\Users\lenovo\OneDrive\Documents\DSA0308'; & 'C:\Users\lenovo\AppData\Local
\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\
launcher' '64098' '--' 'c:\Users\lenovo\OneDrive\Documents\DSA0308\NLP_C02AT1_1.py'

MORPHOLOGICAL ANALYSIS
-----
Word      Root      Suffix    Type      Normalized
-----
connected connect  ed        Inflectional connect
connecting connect ing       Inflectional connect
connection connect ion      Derivational connect

```

**Interpretation:** All three forms are normalized to connect, allowing related forms to be retrieved together.

## Problem 2 – Morphological Parsing for Sentiment Analysis

**Input words:** unhappy, happiness, happily.

| Word      | Prefix | Root/Base | Suffix | Type         | Normalized |
|-----------|--------|-----------|--------|--------------|------------|
| unhappy   | un     | happy     | -      | Derivational | happy      |
| happiness | -      | happy     | ness   | Derivational | happy      |
| happily   | -      | happy     | ly     | Derivational | happy      |

### Python Program:

```

words = ["unhappy", "happiness", "happily"]
def morphological_parse(word):
    if word.startswith("un"):
        prefix = "un"
        root = word[2:]
        suffix = "-"
        transformation = "Derivational"
    elif word.endswith("ness"):
        prefix = "-"
        root = word[:-4]
        suffix = "ness"
        transformation = "Derivational"
    elif word.endswith("ly"):
        prefix = "-"
        root = word[:-2]
        suffix = "ly"
        transformation = "Derivational"
    else:
        prefix = "-"
        root = word
        suffix = "-"
        transformation = "None"
    return prefix, root, suffix, transformation
print("\nMORPHOLOGICAL PARSING")
print("-" * 85)

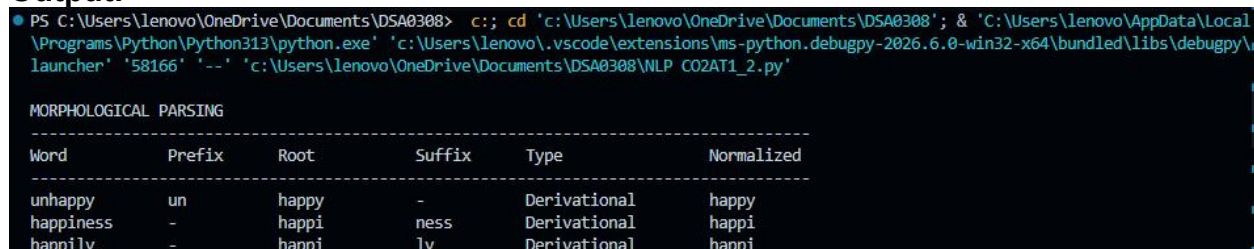
```

```

print(f'{"Word":<15}{ "Prefix":<12}{ "Root":<15}"
      f'{"Suffix":<12}{ "Type":<20}{ "Normalized":<15}')
print("-" * 85)
for word in words:
    prefix, root, suffix, transformation = morphological_parse(word)
    print(f'{"word":<15}{prefix:<12}{root:<15}"
          f'{"suffix:<12}{transformation:<20}{root:<15}')

```

### Output:



```

PS C:\Users\lenovo\OneDrive\Documents\DSA0308> c:: cd 'c:\Users\lenovo\OneDrive\Documents\DSA0308'; & 'C:\Users\lenovo\AppData\Local\
\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\
launcher' '58166' '--' 'c:\Users\lenovo\OneDrive\Documents\DSA0308\NLP_C02AT1_2.py'

MORPHOLOGICAL PARSING
-----
Word      Prefix  Root    Suffix  Type      Normalized
-----
unhappy   un      happy   -        Derivational  happy
happiness -       happi   ness    Derivational  happi
happily   -       happi   ly       Derivational  happi

```

**Interpretation:** The common semantic root is happy. Normalization helps sentiment systems treat related derived forms consistently.

## Problem 3 – Stemming-Based Search Engine Module

**Input words:** played, player, playing.

| Original | Stem | Removed Affix | Transformation | Normalized |
|----------|------|---------------|----------------|------------|
| played   | play | ed            | Inflectional   | play       |
| player   | play | er            | Derivational   | play       |
| playing  | play | ing           | Inflectional   | play       |

### Python Program:

```

words = ["played", "player", "playing"]
def stem_word(word):
    if word.endswith("ing"):
        stem = word[:-3]
        affix = "ing"
        transformation = "Inflectional"
    elif word.endswith("ed"):
        stem = word[:-2]
        affix = "ed"
        transformation = "Inflectional"
    elif word.endswith("er"):
        stem = word[:-2]
        affix = "er"
        transformation = "Derivational"
    else:
        stem = word
        affix = "-"
        transformation = "None"
    return stem, affix, transformation
print("\nSTEMMING ANALYSIS")
print("-" * 85)
print(f'{"Original":<15}{ "Stem":<15}{ "Affix":<12}"
      f'{"Transformation":<20}{ "Normalized":<15}')
print("-" * 85)

```

for word in words:

```
    stem, affix, transformation = stem_word(word)
```

```
    print(f"{word:<15}{stem:<15}{affix:<12}"
```

```
          f"{transformation:<20}{stem:<15}")
```

## Output:

```
PS C:\Users\lenovo\OneDrive\Documents\DSA0308> c;; cd 'c:\Users\lenovo\OneDrive\Documents\DSA0308'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '64151' '--' 'c:\Users\lenovo\OneDrive\Documents\DSA0308\NLP_COZAT1_3.py'
```

```
STEMMING ANALYSIS
```

| Original | Stem | Affix | Transformation | Normalized |
|----------|------|-------|----------------|------------|
| played   | play | ed    | Inflectional   | play       |
| player   | play | er    | Derivational   | play       |
| playing  | play | ing   | Inflectional   | play       |

**Interpretation:** played, player, and playing are normalized to play, improving matching in search and information retrieval.

## Problem 4 – Finite-State Morphological Parsing

**Input words:** writes, writing, written.

**Finite-state model:**

START → ROOT → AFFIX → ACCEPT

For the irregular form written, an irregular transition is used:

START → ROOT → IRREGULAR → ACCEPT

| Word    | State Path                 | Morphology      | Root  | Pattern              | Normalized |
|---------|----------------------------|-----------------|-------|----------------------|------------|
| writes  | q0 → q1 → q2 → q3          | write + s       | write | Regular Inflection   | write      |
| writing | q0 → q1 → q2 → q3          | write + ing     | write | Regular Inflection   | write      |
| written | q0 → q1 → q_irregular → q3 | write → written | write | Irregular Inflection | write      |

## Python Program:

```
words = ["writes", "writing", "written"]
```

```
def finite_state_parser(word):
```

```
    if word == "writes":
```

```
        path = ["q0", "q1", "q2", "q3"]
```

```
        root = "write"
```

```
        morpheme = "write + s"
```

```
        pattern = "Regular Inflection"
```

```
        normalized = "write"
```

```
    elif word == "writing":
```

```
        path = ["q0", "q1", "q2", "q3"]
```

```
        root = "write"
```

```
        morpheme = "write + ing"
```



```

    pattern = "Regular Inflection"
    normalized = "write"
elif word == "written":
    path = ["q0", "q1", "q_irregular", "q3"]
    root = "write"
    morpheme = "write → written"
    pattern = "Irregular Inflection"
    normalized = "write"
else:
    path = ["q0", "q_reject"]
    root = word
    morpheme = "-"
    pattern = "Unknown"
    normalized = word
return path, morpheme, root, pattern, normalized
print("\nFINITE-STATE MORPHOLOGICAL PARSER")
print("-" * 110)
print(f'{"Word":<12}{"State Path":<35}{"Morphology":<20}'
      f'{"Root":<12}{"Pattern":<22}{"Normalized":<12}')
print("-" * 110)
for word in words:
    path, morpheme, root, pattern, normalized = finite_state_parser(word)
    print(f'{"word":<12}' → '.join(path):<35}'
          f'{"morpheme":<20}{"root":<12}'
          f'{"pattern":<22}{"normalized":<12}')

```

### Output:

```

PS C:\Users\lenovo\OneDrive\Documents\DSA0308> c:: cd 'c:\Users\lenovo\OneDrive\Documents\DSA0308'; & 'C:\Users\lenovo\AppData\Local
\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\
launcher' '64283' '--' 'c:\Users\lenovo\OneDrive\Documents\DSA0308\WLP_CO2AT1_4.py'

FINITE-STATE MORPHOLOGICAL PARSER
-----
Word      State Path      Morphology      Root      Pattern      Normalized
-----
writes    q0 → q1 → q2 → q3  write + s      write    Regular Inflection  write
writing   q0 → q1 → q2 → q3  write + ing    write    Regular Inflection  write
written   q0 → q1 → q_irregular → q3  write → written  write    Irregular Inflection  write

```

**Interpretation:** writes and writing follow regular transitions, while written is handled through a special irregular transition. All forms normalize to write.

## Problem 5 – Porter Stemmer-Based Preprocessing

**Input words:** relational, relation, relate.

| Original   | Step   | Rule          | Intermediate Form | Final Stem |
|------------|--------|---------------|-------------------|------------|
| relational | Step 2 | ATIONAL → ATE | relate            | relat      |
| relational | Step 5 | E → ∅         | relat             | relat      |
| relation   | Step 4 | ION removal   | relat             | relat      |
| relate     | Step 5 | E → ∅         | relat             | relat      |

### Python Program:

```

words = ["relational", "relation", "relate"]
def porter_stem(word):

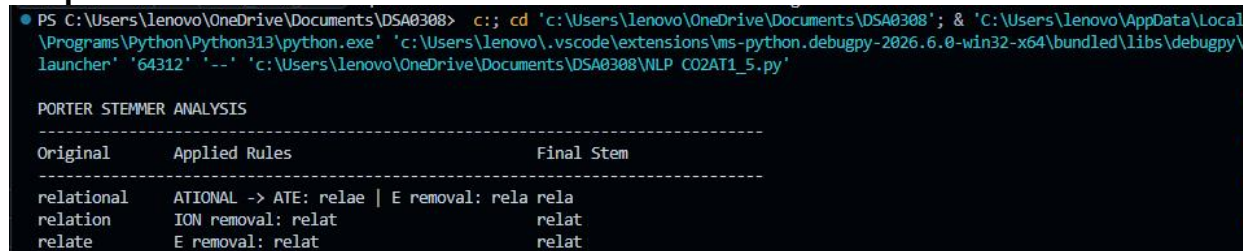
```

```

original = word
steps = []
# Step 2: ational -> ate
if word.endswith("ational"):
    word = word[:-6] + "e"
    steps.append("ATIONAL -> ATE: " + word)
# Step 2: tional -> tion
elif word.endswith("tional"):
    word = word[:-2]
    steps.append("TIONAL -> TION: " + word)
# Step 4: remove ion
if word.endswith("ion"):
    stem = word[:-3]
    if stem.endswith("at"):
        word = stem
        steps.append("ION removal: " + word)
# Step 5: remove final e
if word.endswith("e") and len(word) > 3:
    word = word[:-1]
    steps.append("E removal: " + word)
return original, steps, word
print("\nPORTER STEMMER ANALYSIS")
print("-" * 80)
print(f'Original':<15}{Applied Rules':<40}{Final Stem':<15}")
print("-" * 80)
for word in words:
    original, steps, stem = porter_stem(word)
    rules = " | ".join(steps)
    print(f'original:<15}{rules:<40}{stem:<15}")

```

## Output:



```

PORTER STEMMER ANALYSIS
-----
Original      Applied Rules      Final Stem
-----
relational    ATIONAL -> ATE: rela | E removal: rela rela
relation      ION removal: relat      relat
relate        E removal: relat      relat

```

**Interpretation:** The Porter rules reduce relational, relation, and relate to the computational stem relat.

## Final Comparative Summary

| Q | Input                                   | Main Method                    | Classification                 | Normalized Output |
|---|-----------------------------------------|--------------------------------|--------------------------------|-------------------|
| 1 | connected,<br>connecting,<br>connection | Morphological<br>decomposition | Inflectional +<br>Derivational | connect           |
| 2 | unhappy,<br>happiness,                  | Rule-based<br>parsing          | Derivational                   | happy             |

|   |                              |                      |                             |       |
|---|------------------------------|----------------------|-----------------------------|-------|
|   | happily                      |                      |                             |       |
| 3 | played, player, playing      | Stemming             | Inflectional + Derivational | play  |
| 4 | writes, writing, written     | Finite-state parsing | Regular + Irregular         | write |
| 5 | relational, relation, relate | Porter stemming      | Derivational stemming       | relat |

## Key Concepts

1. **Inflectional morphology:** Changes grammatical form without normally creating a new lexical word, e.g., play → played.
2. **Derivational morphology:** Creates a new lexical form or changes meaning/category, e.g., happy → happiness.
3. **Stemming:** Removes affixes to obtain a computational stem, which may not be a complete English word.
4. **Morphological normalization:** Maps related word forms to a common representation for retrieval.
5. **Finite-state parsing:** Uses states and transitions to recognize roots, suffixes, and irregular morphological patterns.

## Conclusion

The five problems demonstrate different approaches to morphological processing. Rule-based decomposition identifies roots and affixes, stemming reduces grammatical and derivational variations, finite-state parsing represents morphological transitions, and Porter stemming produces compact computational stems. These methods support efficient document indexing, search, sentiment analysis, and general natural language processing.